

LESSON 2 · 00:00

The tilde trick solves constants

Desmos regression can solve for unknown constants even when there is no data set. Put known coordinates into a tiny table, replace equals with tilde, and let Desmos estimate the parameter.

$$y_1 \sim \text{expression with } k$$

f

CHECK

Today: custom regressions, constants, and checks

CORE IDEA

One point can determine one constant

If one unknown constant appears in an equation and one point lies on its graph, a one-row table gives one equation. Desmos can solve that parameter with a custom regression.

x_1	y_1
4	19

CHECK

Point (4, 19) gives $k = 7$

GUIDED EXAMPLE

Exercise 1 · Quadratic constant

The point (3, 26) lies on $y = 2x^2 + kx - 1$.
Find k . Enter 3 and 26 into a table, then type the custom regression.

$$y_1 \sim 2x_1^2 + kx_1 - 1$$

f

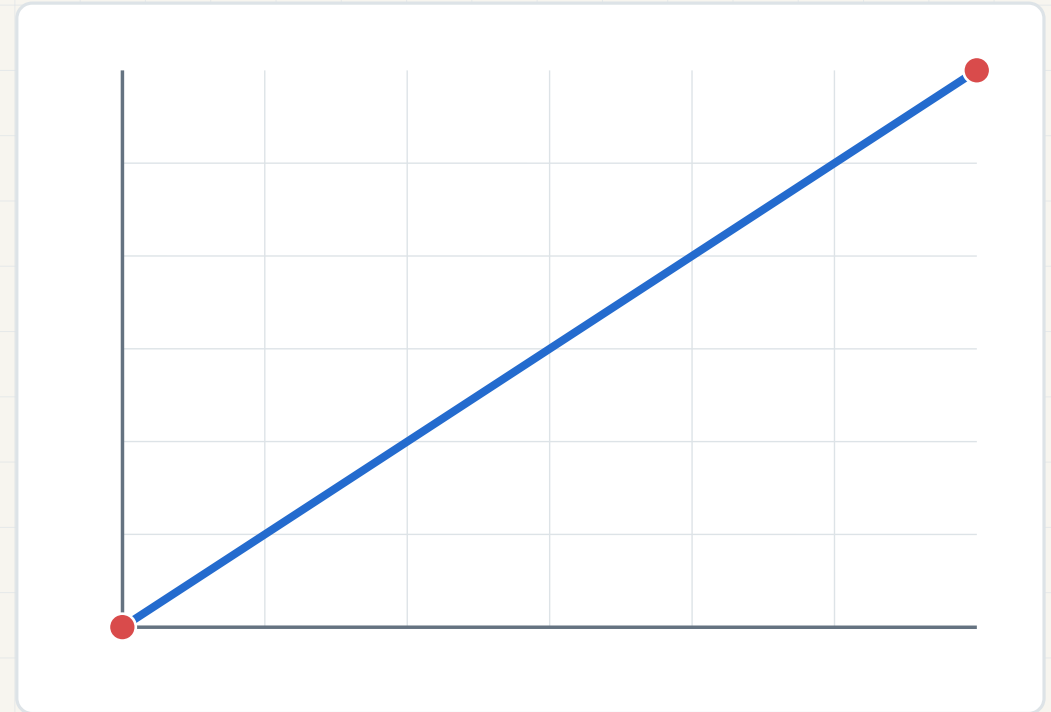
CHECK

$$26 = 18 + 3k - 1, \text{ so } k = 3$$

TWO POINTS, TWO PARAMETERS

Exercise 2 · Two unknowns

A line $y = ax + b$ passes through $(2, 11)$ and $(5, 23)$. Put both points in the table and regress y_1 on $ax_1 + b$.



CHECK

 $a = 4$ and $b = 3$

BEFORE YOU TYPE

The information-count rule

Usually, n independent unknown parameters require at least n independent conditions. One point cannot uniquely determine both slope and intercept. Two identical x -values also fail to determine a nonvertical line.

$$\text{unknowns} \leq \text{independent conditions}$$

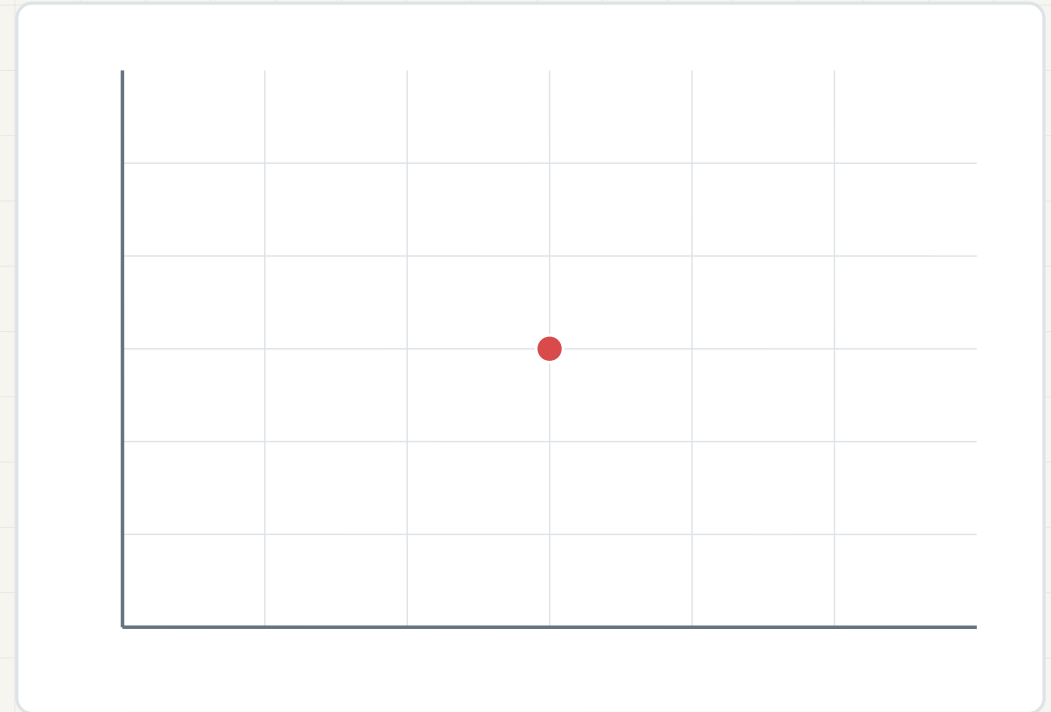
CHECK

Count information, not rows

TRY IT · PAUSE 40 SECONDS

Exercise 3 · Circle radius

The circle $x^2 + y^2 = r^2$ passes through (5, 12). Use a custom regression to find its positive radius.



CHECK **$25 + 144 = 169$, so $r = 13$**

SAT-STYLE MODEL

Exercise 4 · Exponential parameter

The function $f(x) = 80 \cdot b^x$ passes through $(3, 270)$. Find b . Enter the point and fit the unknown base.

$$y_1 \sim 80b^{(x_1)}$$

f

CHECK

$$b^3 = 27/8, \text{ so } b = 1.5$$

VERTEX FORM

Transformations with an unknown

A parabola $y = (x - h)^2 + 4$ has x-intercept -1 . Find h . A point on the graph is $(-1, 0)$, so fit h from that one row.

$$y_1 \sim (x_1 - h)^2 + 4$$

f

CHECK

 $0 = (-1 - h)^2 + 4$ has no real solution

VERIFY THE OUTPUT

Catch impossible conditions

Desmos regression works over real numbers. Since a square is never negative, $(-1 - h)^2 + 4$ cannot equal zero. If a problem's condition is impossible, do not force a decimal answer.

$$(x - h)^2 \geq 0 \Rightarrow y \geq 4$$

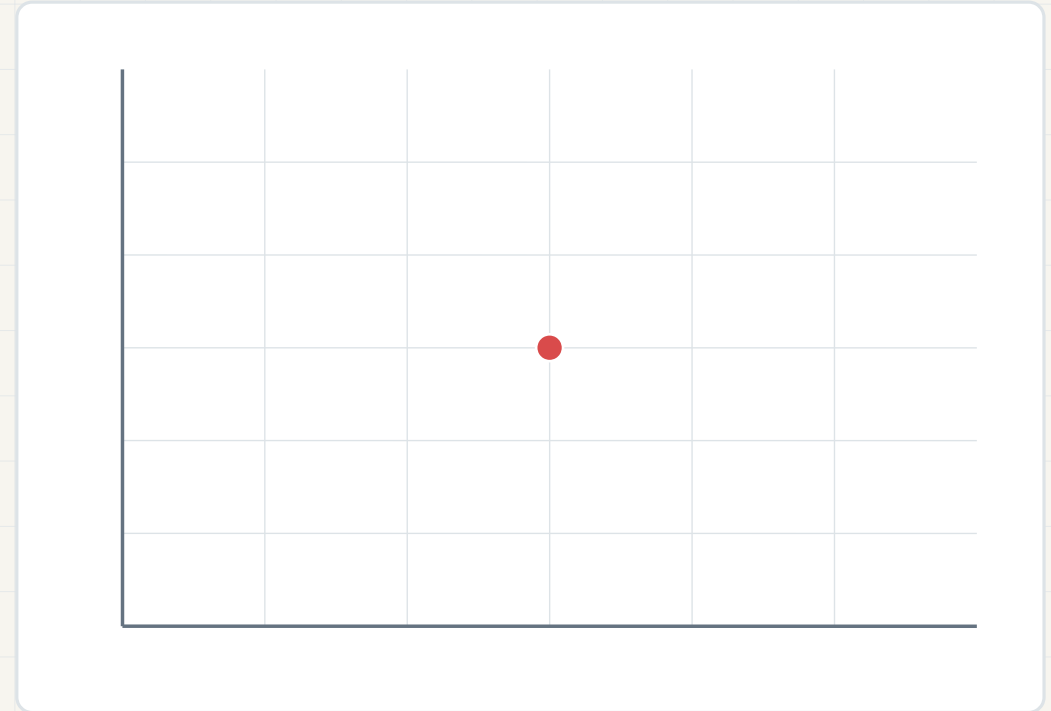
CHECK

No real value of h

INDEPENDENT PRACTICE

Exercise 5 · Corrected vertex problem

A parabola $y = (x - h)^2 - 4$ has x-intercept -1 and vertex to the right of the y-axis. Find h .



CHECK

 $h = 1$ or -3 ; condition $h > 0$ selects $h = 1$

MULTI-CONDITION CHALLENGE

Exercise 6 · Parameter in a system

The lines $y = kx + 2$ and $y = -2x + 17$ intersect where $x = 3$. Find k , then find the intersection point.

$$3k + 2 = -2(3) + 17$$

f

CHECK **$k = 3$; intersection is $(3, 11)$**

ACCURACY CHECK

Avoid four common mistakes

Use tilde for regression. Use x_1 and y_1 from the table, not plain x and y . Use enough independent conditions. Finally, substitute every fitted value into the original statement.

~ → **subscripts** → **enough data** → **verify**

CHECK

Desmos is fast; verification makes it reliable

PAUSE 50 SECONDS

Final challenge

The quadratic $y = ax^2 + bx + 5$ passes through $(1, 12)$ and $(3, 44)$. Find $a + b$.

x_1	y_1
1	12
3	44

CHECK **$a = 3, b = 4, \text{ so } a + b = 7$**

EXAM-DAY ROUTINE

Lesson 2 recap

Translate each condition into a point or equation. Count unknowns. Build the smallest useful table. Fit with a tilde. Then substitute back into the original problem.

**TRANSLATE → FIT →
VERIFY**

f

CHECK

You now have a repeatable method for hard constant questions